

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Text Structure and Purpose	Hard

Question

Mathematician Claude Shannon is widely regarded as a foundational figure in information theory. His most important paper, “A Mathematical Theory of Communication,” published in 1948 when he was employed at Bell Labs, utilized a concept called a “binary digit” (shortened to “bit”) to measure the amount of information in any signal and determine the fastest rate at which information could be transmitted while still being reliably decipherable. Robert Gallagher, one of Shannon’s colleagues, said that the bit was “[Shannon’s] discovery, and from it the whole communications revolution has sprung.”

Which choice best describes the overall structure of the text?

Answer

- A. It presents a theoretical concept, illustrates how the name of the concept has changed, and shows how the name has entered common usage.
- B. It introduces a respected researcher, describes an aspect of his work, and suggests why the work is historically significant.
- C. It names the company where an important mathematician worked, details the mathematician’s career at the company, and provides an example of the recognition he received there.
- D. It mentions a paper, offers a summary of the paper’s findings, and presents a researcher’s commentary on the paper.

Correct Answer: B

Rationale

Choice B is the best answer. The text starts with a general statement that introduces Shannon, then describes a specific contribution from one of his papers, then provides a quote that illustrates just how important this contribution was.

Choice A is incorrect. This isn’t the overall structure. The text only mentions very briefly, in parentheses, that “binary digit” was shortened to “bit.” It doesn’t go into detail about this name change, and it doesn’t discuss any “common usage” of the name at all. Choice C is incorrect. This isn’t the overall structure. Shannon’s employment at Bell Labs is only mentioned once, very briefly: the text never goes into detail about his career there, and it never mentions any recognition he received there. Choice D is incorrect. This is too narrow. Overall, the text is about Shannon’s importance in his field, not just this one paper of his.